**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 071 · Subprograms and Arrays

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. When might a procedure be suitable?

Q2. In OCR ERL, s = 'VARIABLE'. What is s.substring(1,3)?
A. VAR
B. RIA
C. ARI
D. RIAB

Q3. “Failure of any one workstation cable necessarily disables all other star-network links.” Correct this. Explain why.

Q4. A network must cope if one link fails. “A mesh has only one path between any two devices.” Correct this. How could it keep working?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Subprogram — A procedure or function defines reusable instructions.

Array parameter — An array can be supplied to a subprogram so values can be processed.

Return value — A function passes a result back to the code that called it.

Testing — Test a subprogram using several arrays including boundary cases.

Algorithm — A defined set of processing instructions.

Testing — Checking expected outcomes against actual program behaviour.

## New learning

Subprogram: A procedure or function defines reusable instructions. In the score-array helper, identify where this technique applies.

Array parameter: An array can be supplied to a subprogram so values can be processed. In the library stock function, identify where this technique applies.

Return value: A function passes a result back to the code that called it. In the sports result processor, identify where this technique applies.

Testing: Test a subprogram using several arrays including boundary cases. In the classroom rating program, identify where this technique applies.

## Worked example

A score-array helper uses subprogram and array parameter. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. Subprogram [1]; A procedure or function defines reusable instructions. [1].

3. Array parameter [1]; An array can be supplied to a subprogram so values can be processed. [1].

4. Each feature needs a correct technical fact plus an explanation.

# We Do · Practise with guidance

### Guided 1

A library stock function needs subprogram. Explain the purpose. [2 marks]

Worked method: Subprogram [1]; A procedure or function defines reusable instructions. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Guided 2

Explain array parameter for a sports result processor. [2 marks]

Worked method: Array parameter [1]; An array can be supplied to a subprogram so values can be processed. [1].

## Guided 3

When programming a classroom rating program, explain why return value and testing matter. [4 marks]

Worked method: Return value [1]; A function passes a result back to the code that called it. [1]. Testing [1]; Test a subprogram using several arrays including boundary cases. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Finish the comparison that lets the function return the largest array value.

```
function largest(values)
    biggest = values[0]
    for i = 1 to 2
        if ... then
            biggest = values[i]
        endif
    next i
    return biggest
endfunction
array disks = [32,128,64]
print(largest(disks))
```

write

### Question 1

State TWO relevant ideas or operations from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

### Question 2

Explain array parameter and return value in a temperature array report. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

### Question 3

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A programmer makes a competition leaderboard. Explain the roles of subprogram, return value and testing. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

## Exit Ticket · Answer before checking

A procedure or function defines reusable instructions.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. For a named reusable task such as displaying a menu when no function result is required.

Q2. C — ARI

Q3. If the central switch fails, devices using it lose contact. A broken cable usually affects just one workstation.

Q4. A mesh has other paths if some links fail. More links add cost and complexity. Check the paths that remain.

## I Do / We Do pages · Guided practice

Q1. Subprogram [1]; A procedure or function defines reusable instructions. [1].

Q2. Array parameter [1]; An array can be supplied to a subprogram so values can be processed. [1].

Q3. Return value [1]; A function passes a result back to the code that called it. [1]. Testing [1]; Test a subprogram using several arrays including boundary cases. [1].

## You Do pages · Code First

```
function largest(values)
    biggest = values[0]
    for i = 1 to 2
        if values[i] > biggest then
            biggest = values[i]
        endif
    next i
    return biggest
endfunction
array disks = [32,128,64]
print(largest(disks))
```

Expected result: 128

## You Do pages · Independent questions

Q1. Subprogram [1]; Array parameter [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Q2. Array parameter [1]; An array can be supplied to a subprogram so values can be processed. [1]. Return value [1]; A function passes a result back to the code that called it. [1].

Q3. Subprogram [1]; A procedure or function defines reusable instructions. [1]. Return value [1]; A function passes a result back to the code that called it. [1]. Testing [1]; Test a subprogram using several arrays including boundary cases. [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**